

Ph.D. Exam on Integration Theory

May 9, 1994

I. State the following theorems:

- 1) The Uniform Boundedness theorem
- 2) The Hahn–Banach theorem
- 3) The Fatou Lemma
- 4) The Lebesgue dominated convergence theorem
- 5) The Vitali theorem
- 6) The Radon–Nikodym theorem
- 7) The Egorov theorem
- 8) The Fubini theorem
- 9) The dual of L^p ($1 \leq p < \infty$)
- 10) The Lusin theorem
- 11) The Riesz representation theorem in locally compact spaces
- 12) The Lebesgue decomposition theorem

II. Prove one of the theorems 8 or 9.

III. Solve the following problems:

(X, Σ, μ) is a measure space, E is a Banach space.

- 1) Let (f_n) be a sequence of $L_E^1(\mu)$ such that $\int |f_n| d\mu < \frac{1}{n^2}$ for every n . Prove or disprove the pointwise convergence and the convergence in the mean of the series $\sum f_n$.
- 2) Assume μ is finite and let R be a ring generating Σ . Define $\rho(A, B) = \mu(A \Delta B)$ for $A, B \in \Sigma$. Prove that ρ is a semi-distance and that R is dense in Σ for ρ .
- 3) Let $X = \mathbb{R}$ and μ the Lebesgue measure. Let $f \in \mathcal{L}_E^1(\mu)$ such that $\int_0^x f d\mu = 0$ for every $x \in \mathbb{R}$ (if $x < 0$, $\int_0^x = -\int_x^0$). Prove that $f = 0, \mu$ -a.e.
- 4) Let μ, ν be finite, σ -additive measures on Σ , such that $\mu \ll \nu$ and $\nu \ll \mu$. What can be said about the relationship of the Radon–Nikodym densities of one measure with respect to the other.

- 5) Let (x_n) be a sequence in the Banach space E . Assume $x_n \rightarrow x$ weakly in E . Show that x is the only weak limit of the sequence (x_n) and that $\sup_n \|x_n\| < \infty$. (Hint: Consider the x_n as elements of the bidual E^{**}).
- 6) Let $X = \mathbb{R}$ and μ the Lebesgue measure. Let $f \in L^1(\mu)$ and for each $t \in \mathbb{R}$ define $f_t(x) = f(x - t)$. Show that the mapping $t \mapsto f_t$ is a continuous map from $\mathbb{R}$ into $L^1(\mu)$. (Hint: prove it first for continuous functions with compact support).